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Overlay target and overlay method

Abstract

An overlay target includes a plurality of working zones, a plurality of holes in each of the working zones, and a first layer filling in the plurality of holes. The plurality of holes are not filled up by the first layer, and a plurality of spaces are reserved in the plurality of holes.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims the priority benefit of Taiwan application serial no. 111135839, filed on Sep. 22, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.
BACKGROUND
Technical Field
(2) The disclosure relates to a semiconductor device and a manufacturing method thereof, and in particular relates to an overlay target and an overlay method for an overlay mark.
DESCRIPTION OF RELATED ART
(3) A lithography process is a patterning technology widely used in semiconductor manufacturing process. The lithography process is usually aligned by overlay marks. However, if a blanket-type metal layer is to be patterned, it is difficult to calculate the correct overlay error because the metal layer is opaque.
SUMMARY
(4) The disclosure provides an overlay target, which may be used to accurately measure overlay errors.
(5) An overlay target of an embodiment of the disclosure includes multiple working zones, multiple holes in each of the working zones, and a first layer filling in the holes. The holes are not filled up

by the first layer, and multiple spaces are reserved.
(6) An overlay method of an embodiment of the disclosure is used to determine an overlay error of a device structure formed in a semiconductor manufacturing process, including the following operation. A first overlay target is formed, including the following operation. Multiple working zones are provided, and multiple holes are formed in a first sub-zone of each of the working zones. A first layer is formed in the holes, in which the holes are not filled up by the first layer, and multiple spaces are reserved. The spaces have a first diameter and a second diameter greater than the first diameter. Multiple midpoints of the second diameters of the spaces are taken as multiple calibration center points of the holes. A second overlay target is formed, including forming multiple line segments in a second sub-zone of each of the working zones. The overlay error of the device structure is determined by using the calibration center points and multiple centers of the line segments.
(7) Based on the above, the overlay target of the overlay mark according to the embodiment of the disclosure may overcome the problem that the calculation of the overlay error is difficult because the previous layer is a blanket-type opaque material, and may accurately calculate the correct overlay error.

Description

BRIEF DESCRIPTION OF THE DRAWING
(1) FIG. **1** is a top view of an overlay mark in accordance with an embodiment of the disclosure.
(2) FIG. **2** is a waveform diagram of the third working zone of the first overlay target of FIG. **1**.
(3) FIG. **3**A is a partial enlarged view of a unit of the first overlay target of FIG. **1**.
(4) FIG. **3**B is a schematic view of the space within the hole of FIG. **3**A.
(5) FIG. **3**C is a schematic view of a dummy hole reconstructed according to the calibration center point and the second diameter of FIG. **3**B.
(6) FIG. **4** is a cross-sectional schematic view of a unit of a first overlay target in an overlay mark zone according to an embodiment of the disclosure.
(7) FIG. **5** is a top view of the units of the first overlay target at various locations on the wafer.
DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS
(8) FIG. **1** is a top view of an overlay mark in accordance with an embodiment of the disclosure. FIG. **2** is a waveform diagram of the third working zone of the first overlay target of FIG. **1**. FIG. **3**A is a partial enlarged view of a unit of the first overlay target of FIG. **1**. FIG. **3**B is a schematic view of the space within the hole of FIG. **3**A. FIG. **3**C is a schematic view of a dummy hole reconstructed according to the calibration center point and the second diameter of FIG. **3**B.
(9) Referring to FIG. **1**, an overlay mark **100** includes multiple working zones **10**, a first overlay target **100**A, and a second overlay target **100**B. Each of the working zones **10** includes a first sub-zone **10**A and a second sub-zone **10**B. In this embodiment, the working zones **10** include, for example, a first working zone **11**, a second working zone **12**, a third working zone **13**, and a fourth working zone **14**. The width, length, and area of the first working zone **11**, the second working zone **12**, the third working zone **13**, and the fourth working zone **14** may be approximately equal, and the configurations are rotationally symmetric. The first working zone **11** includes a first sub-zone **11**A and a second sub-zone **11**B. The second working zone **12** includes a first sub-zone **12**A and a second sub-zone **12**B. The third working zone **13** includes a first sub-zone **13**A and a second sub-zone **13**B. The fourth working zone **14** includes a first sub-zone **14**A and a second sub-zone **14**B.
(10) Referring to FIG. **1**, a first overlay target **100**A may be used for a previous layer of a semiconductor device. The second overlay target **100**B may be used for the present layer of the semiconductor device. The first overlay target **100**A of the overlay mark **100** is adjacent to the

second overlay target **100**B. The first overlay target **100**A is disposed in the first sub-zone **10**A. The second overlay target **100**B is disposed in the second sub-zone **10**B.

(11) Referring to FIG. **1**, multiple second sub-zones **10**B surround the center point **10**C of the overlay mark **100**, and the configuration thereof is rotationally symmetric. The width, length, and area of the second sub-zones **10**B may be approximately equal. The second overlay target **100**B includes multiple units **50**B. Each unit **50**B is a line segment **20**. In some embodiments, the line segments **20** may be rectangular. The rectangular line segments **20** are approximately equal in width, approximately equal in length, approximately equal in area, and are parallel to each other. The widths WG**2** of the gaps **20**G between the line segments **20** may be approximately equal. However, the disclosure is not limited thereto.

(12) Referring to FIG. **1**, the line segments **20** of the second sub-zone **11**B of the first working zone **11** and the line segments **20** of the second sub-zone **13**B of the third working zone **13** extend along the D**2** direction and are arranged along the direction D**1**. The line segments **20** of the second sub-zone **12**B of the second working zone **12** and the line segments **20** of the second sub-zone **14**B of the fourth working zone **14** extend along the direction D**1** and are arranged along the direction D**2**. The direction D**1** and the direction D**2** are perpendicular to each other. In other words, the line segments **20** in the second sub-zone **11**B are parallel to each other and parallel to the line segments **20** of the second sub-zone **13**B. The line segments **20** in the second sub-zone **12**B are parallel to each other and parallel to the line segments **20** of the second sub-zone **14**B.

(13) In some embodiments, the line segments **20** in the second sub-zones **11**B and **13**B are symmetrical to the center point **10**C. The line segments **20** in the second sub-zones **12**B and **14**B are symmetrical to the center point **10**C. In other words, the length, width, number, arrangement of the line segments **20** and the width WG**2** of the gap **20**G in the second sub-zone **11**B are the same as the length, width, number, arrangement of the line segments **20** and the width WG**2** of the gap **20**G in the second sub-zone **13**B. The length, width, number, arrangement of the line segments **20** and the width WG**2** of the gap **20**G in the second sub-zone **12**B are the same as the length, width, number, arrangement of the line segments **20** and the width WG**2** of the gap **20**G in the second sub-zone **14**B. However, the disclosure is not limited thereto. The units **50**B in each of the sub-zones **10**B may be arranged in other shapes, lengths, widths, and numbers, and the widths WG**2** of the gaps **20**G may be equal or unequal.

(14) Referring to FIG. **1**, the shape of the first overlay target **100**A is different from the shape of the second overlay target **100**B. The first overlay target **100**A includes multiple first sub-zones **10**A, and multiple holes **30** and a first layer **32** in each of the first sub-zones **10**A.

(15) The first sub-zones **10**A of the first overlay target **100**A surround multiple second sub-zones **10**B. In some embodiments, the configuration of the first sub-zones **10**A is rotationally symmetric. For example, the first sub-zones **10**A of the first overlay target **100**A include first sub-zones **11**A, **12**A, **13**A, and **14**A. The first sub-zone **11**A of the first working zone **11** is longitudinally adjacent to the second sub-zone **11**B. The first sub-zone **12**A of the second working zone **12** is laterally adjacent to the second sub-zone **12**B. The first sub-zone **13**A of the third working zone **13** is longitudinally adjacent to the second sub-zone **13**B. The first sub-zone **14**A of the fourth working zone **14** is laterally adjacent to the second sub-zone **14**B. The width, length, and area of the first sub-zones **11**A to **14**A may be approximately equal.

(16) Referring to FIG. **1**, the first overlay target **100**A includes multiple units **50**A. A unit **50**A is a hole **30**. In some embodiments, the holes **30** in the same working zone **10** are in a regular pattern, with approximately the same size and approximately the same shape. In some embodiments, the diameter d**2** of the hole **30** is, for example, 0.6 micrometers to 1.5 micrometers. The widths WG**1** of the gaps **30**G between the holes **30** are approximately equal. The holes **30** may be arranged in an array. For example, in some embodiments, the holes **30** in the first sub-zone **11**A of the first working zone **11** and in the first sub-zone **13**A of the third working zone **13** may be respectively arranged in a 4×2 array, and the holes **30** in the first sub-zone **12**A of the second working zone **12**

and in the first sub-zone **14**A of the fourth working zone **14** may be respectively arranged in a 2×4 array. However, the embodiments of the disclosure are not limited thereto. In some embodiments of the disclosure, the holes **30** in each column of the first working zone **11** and the third working zone **13** correspond to a single line segment **20**, and the holes **30** in each row of the second working zone **12** and the fourth working zone **14** correspond to a single line segment **20**. However, the embodiments of the disclosure are not limited thereto.

(17) FIG. **4** is a cross-sectional schematic view of a unit of a first overlay target in an overlay mark zone according to an embodiment of the disclosure.

(18) Referring to FIG. **4**, multiple holes **30** are formed in a second layer **36**. In some embodiments, the holes **30** extend from the upper surface of the second layer **36** to the lower surface, but do not penetrate the second layer **36**. In other embodiments, the holes **30** may extend from the upper surface of the second layer **36** to the lower surface and penetrate the second layer **36** (not shown). The second layer **36** may be a dielectric material on a semiconductor substrate, such as silicon oxide, silicon nitride, aluminum oxide, or a low-k (low dielectric constant) dielectric material.

(19) Referring to FIG. **1**, FIG. **3**A, and FIG. **4**, the first layer **32** of the first overlay target **100**A covers the second layer **36** and fills the holes **30**. The first layer **32** is deposited by a physical vapor deposition method, such as sputtering, therefore, the first layer **32** may also be called a deposition layer or a blanket layer. In some embodiments, the first layer **32** is an opaque layer. The opaque layer is, for example, a metal layer. The metal layer is, for example, Ta, Al, Ir, or a combination thereof. The metal layer may be a single layer or multiple layers.

(20) Referring to FIG. **4**, in the disclosure, the holes **30** are not filled up by the first layer **32**, and multiple spaces **34** are reserved. Therefore, the maximum thickness of the first layer **32** (e.g., the thickness T**1**) is less than the radius of the hole **30** (i.e., half of the diameter d**3**). Furthermore, the thickness of the first layer **32** on the sidewall of the hole **30** is not the same due to the characteristics of the manufacturing process machine. For example, referring to FIG. **4**, in some embodiments, the first layer **32** is formed by physical vapor deposition (PVD), and is affected by the PVD shadowing effect, such that the first layer **32** on the two side walls of the hole **30** has different thicknesses T**1** and T**2**. The shapes of the spaces **34** are different from the shapes of the holes **30** due to the different thicknesses of the first layer **32** on the side walls of the holes **30**. In some embodiments, the holes **30** are centrally symmetric, such as circular, and the spaces **34** are asymmetric, irregular circles, such as off-axis football shapes, as shown in FIG. **1** and FIG. **3**A. In FIG. **1**, the spaces **34** have a regular pattern, however, the shapes of the spaces **34** may be different and irregular due to manufacturing processing factors.

(21) Referring to FIG. **3**A and FIG. **3**B, the hole **30** has a diameter d**3**. The diameters d**3** of the holes **30** in the same working zone **10** are approximately the same or similar. However, each of the spaces **34** reserved in the hole **30** has a first diameter d**1** and a second diameter d**2** larger than the first diameter d**1** for reasons explained below.

(22) FIG. **5** is a top view of the units of the first overlay target at various locations on the wafer.

(23) Referring to FIG. **5**, since the distances between different positions of a wafer **1000** and the target are different, while also affected by the PVD shadowing effect, the thickness of the first layer **32** formed in the holes **30** in a specific direction (e.g., the direction in which the first diameter d**1** extends) is not uniform, while the other direction (e.g., the direction in which the second diameter d**2** extends) is less affected by the shadowing effect, so that the thickness in the other direction is approximately equal. Moreover, the thickness of the first layer **32** formed in the holes **30** of the wafer **1000** has different distributions according to the positions of the holes **30**, so that the extending direction of the second diameter d**2** of the spaces **34** left by the holes **30** at different positions are different. Therefore, the units **50**A of the first overlay target **100**A of the disclosure may be disposed on various positions of the wafer **1000**. For the sake of brevity, the hole **30** and the space **34** in FIG. **3**A are used for illustration, but are not intended to limit the disclosure. The hole **30** and the space **34** in FIG. **3**A are, for example, the hole **30** at the position L**1** in FIG. **5**.

(24) Referring to FIG. **3**A and FIG. **3**B, the first diameter d**1** of the space **34** reserved in the hole **30** is parallel to the direction D**1**, and the second diameter d**2** is parallel to the direction D**2**. The first diameter d**1** is the largest diameter of the space **34** in the direction D**1**, and the second diameter d**2** is the largest diameter of the space **34** in the direction D**2**.
(25) Referring to FIG. **3**A and FIG. **3**B, since the hole **30** is centrally symmetric and the space **34** is asymmetric, the midpoint O**2** of the space **34** in the direction D**1** does not coincide with the center point O**1** of the hole **30** in the direction D**1**. Referring to FIG. **1**, FIG. **2**, and FIG. **3**A, an optical signal **30**S is the actual optical signal of the hole **30** in the third working zone **13** in the direction D**1** (if the first layer **32** is transparent), the center point O**1** of the hole **30** in the direction D**1** should correspond to the peak P**1** of the optical signal **30**S. However, since the first layer **32** is opaque, the optical signal **30**S cannot be obtained. The obtained optical signal **34**S is the optical signal in the direction D**1** actually measured, which is the optical signal in the direction D**1** from the space **34**. The peak P**2** of the light signal **34**S corresponds to the midpoint O**2** of the space **34** in the direction D**1**. The peak P**2** of the optical signal **34**S does not coincide with the peak P**1** of the optical signal **30**S, but has an offset. Therefore, the disclosure does not use the midpoint O**2** of the space **34** in the direction D to calculate the overlay error in the direction D**1**, such as the X direction, but uses the midpoint O**1**′ of the space **34** in the direction D**2** as the calibration center point O**1**′, as detailed below.
(26) Referring to FIG. **3**A to FIG. **3**C, in some embodiments, the thickness of the first layer **32** at the positions E**1** and E**2** of the hole **30**, that is, the two ends of the second diameter d**2** of the space **34**, is the least affected by the shadowing effect, so the thickness difference is very small, even almost equal. Therefore, the midpoint O**1**′ of the space **34** in the direction D**2** is very close to the center point O**1** of the hole **30**, or even coincides with the center point O**1** of the hole **30**.
(27) Therefore, the overlay method of an embodiment of the disclosure used to determine an overlay error of a device structure formed in a semiconductor manufacturing process may obtain information of the second diameter d**2** of the space **34** in the direction D**2** based on the optical signal in the direction D**2**, and may use the midpoint O**1**′ of the second diameter d**2** as the calibration center point O**1**′. Taking the calibration center point O**1**′ as the center of the circle and ½ of the second diameter d**2** as the radius, a dummy hole **30′** may be reconstructed and obtained, as shown in FIG. **3**C. The obtained dummy hole **30′** is symmetric and a regular circle. The obtained calibration center point O**1**′ of the dummy hole **30′** is very close to the center point O**1** of the hole **30**, and may even coincide with the center point O**1** of the hole **30**. Also, the obtained dummy hole **30′** may have the same or similar shape as the hole **30**. In addition, compared to the first diameter d**1** of the space **34**, the obtained second diameter d**2** of the dummy hole **30′** is closer to the diameter d**3** of the hole **30**. In other words, the converted optical signal **30**'S obtained by the dummy holes **30′** is very close to the optical signal **30**S of FIG. **2**.
(28) Afterwards, the overlay error of the device structure may be determined by using multiple calibration center points O**1**′ of each of the working zones **10** and multiple centers of the line segments. For example, the average position Px′ of the previous layer in the X direction may be determined by the calibration center points O**1**′ of the dummy holes **30′** in the first working zone **11** and the third working zone **13**. The average position Px of the present layer in the X direction may be determined by the centers of the line segments **20** of the first working zone **11** and the third working zone **13**. Therefore, the overlay error ΔX between the present layer and the previous layer in the X direction may be obtained from the difference between the average position Px and the average position Px′.
(29) In the second working zone **12** and the fourth working zone **14** of FIG. **1**, the second diameter d**2** of the space **34** in the hole **30** is in the D**2** direction. Obtaining the midpoint O**1**′ of the second diameter d**2** according to the abovementioned method, and using the midpoint O**1**′ as the calibration center point O**1**′, the average position Py′ of the previous layer in the Y direction may be determined. The average position Py of the present layer in the Y direction may be determined by

the line segments **20** of the first working zone **11** and the third working zone **13**. Therefore, the overlay error ΔY between the present layer and the previous layer in the Y direction may be obtained from the difference between the average position Py and the average position Py′. The obtained overlay errors (ΔX, AY) may be immediately fed back to the scanning exposure machine before the etching process, therefore there is need to etch and pattern the first layer **32** to know the result of the overlay. In addition, the disclosure may collect related data such as the offset between the overlay errors (ΔX, AY) in the lithography stage and the overlay errors obtained after etching, to serve as the data of the library of lithograph modeling recipe. In this way, in the lithography stage, the overlay result after etching may be simulated according to the obtained overlay result at the lithography stage.

(30) In other words, by designing the first overlay target for the previous layer as holes through the embodiment of the disclosure, and by filling the holes with the first layer while also reserving spaces without filling up the holes, it is possible to calculate the correct position of the previous layer and then to calculate a substantially correct overlay error as early as in the lithography stage, and to infer the overlay error after etching from the overlay error at the lithography stage. Therefore, the overlay mark of the embodiment of the disclosure may be used to accurately measure the overlay error in the lithography stage, so as to avoid the inability to measure and calculate the overlay error because the previous layer is opaque.

Claims

1. An overlay method, used to determine an overlay error of a device structure formed in a semiconductor manufacturing process, comprising: forming a first overlay target, comprising: providing a plurality of working zones; forming a plurality of holes in a first sub-zone of each of the working zones; and forming a first layer in the holes, wherein the holes are not filled up by the first layer, and a plurality of spaces are reserved, the spaces have a plurality of first diameters and a plurality of second diameters greater than the first diameters; taking a plurality of midpoints of the second diameters of the spaces as a plurality of calibration center points; forming a second overlay target, comprising forming a plurality of line segments in a second sub-zone of each of the working zones; and determining the overlay error of the device structure by using the calibration center points and a plurality of centers of the line segments.
2. The overlay method according to claim 1, wherein the first layer is an opaque layer.
3. The overlay method according to claim 1, wherein the opaque layer comprises a metal layer.
4. The overlay method according to claim 1, wherein the first layer is a deposition layer formed on a second layer and filled in a plurality of openings of the second layer.
5. The overlay method according to claim 4, further comprising that the spaces are asymmetric.
6. The overlay method according to claim 1, wherein a maximum thickness of the first layer in the holes is less than a radius of the holes.
7. The overlay method according to claim 1, wherein a thickness of the first layer in the holes is not uniform.
8. The overlay method according to claim 1, wherein shapes of the spaces are different from shapes of the holes.
9. The overlay method according to claim 8, wherein the holes are circular and the spaces are off-axis football-shaped.